



AI + Workforce Development

Provost Muscat-Back to the Bay 2025 Keynote-One-page summary for faculty

Why this matters now

13%

Demographic cliff

Fewer high school graduates nationally by 2041 **13%** — about 1% every year — with no good replacement revenue as enrollment falls.

\$16K

Debt & value

CSUEB graduates leave with about **\$16,000** in debt, repaid over roughly 8 years, while public confidence in college's value keeps eroding.



Entry-level, redefined

AI is redefining entry-level work **right now**: graduates will be expected to handle complex, judgment-based tasks from day one.

Two gaps to close

The skills gap

What students know at graduation vs. what employers need. **AI is closing it** — advanced simulations, market analysis in minutes, instant first drafts.

The experience gap

Doing the work amid people, deadlines, and shifting priorities. AI can draft the care plan; **only a human** sits with the family in crisis.

A three-part upskilling framework

AI Fluency

Understand how AI works, what it can and **cannot** do, and how to apply it responsibly — always with human verification.

Human-Centric Skills

Critical thinking, communication, collaboration, ethical judgment, curiosity, and relational intelligence — the advantages AI cannot replace.

Agility

Adapt quickly when conditions change: reframe the project, pivot with confidence, and treat change as an opening for innovation.

Bridging to the workforce

Experiential learning is the bridge

Internships, community projects, and the campus as a living lab — built into every program and every year. Graduates leave with a **portfolio**, not just a transcript.

Agentic AI is arriving

AI that plans and executes multi-step work with minimal supervision. Teach students to **lead** agents: set goals, monitor, quality-check, and override.



AI PLUS: it is AI plus humans

AI brings speed, scale, and pattern recognition. Humans bring interpretation, relationships, ethical judgment, and creative synthesis.

The Provost's two asks

Where could AI serve as an accelerant for deeper human interpretation in the work that you do? And: **Where could *not* using AI be an intentional choice that itself teaches discipline, focus, and originality?**

Source: <https://www.csueastbay.edu/oaa/files/docs/provost-muscat-b2b-keynote.pdf>

Made with R and Quarto · Department of Statistics and Data Science · Cal State East Bay



Faculty Checklist

Part A: Getting familiar with AI yourself: 10 items

Before asking students to work with AI, build your own comfort — work down this list at your own pace.

Fluency **Human-Centric** **Agility** — each item is tagged to the Provost's three pillars.

☐ **Fluency** **Try two tools hands-on.** Use two AI tools for a real task this term — the university ChatGPT account, Microsoft Copilot, or Google Gemini — not a demo.

☐ **Fluency** **Grade your own assignment.** Run one of your own assignments through AI. Note what it gets right, what it fabricates, and where students would stumble.

☐ **Fluency** **Practice iterative prompting.** Give context, specify the format, request revisions. Keep a short log of prompts that worked and why.

☐ **Fluency** **Learn the failure modes.** Hallucinated citations, confident wrong math, hidden assumptions — build a personal verification habit you can teach.

☐ **Fluency** **Know the privacy rules.** Never paste student data, unpublished research, or personnel information into public tools. Know what campus licenses.

☐ **Fluency** **Read the campus AI policy.** Read it end to end — plus your department's expectations for disclosure and attribution.

☐ **Fluency** **Attend one AI teaching workshop.** A Faculty Development session, the Back to the Bay series, or a workshop in your discipline.

☐ **Fluency** **Read one substantial piece.** An EDUCAUSE or discipline-journal article on teaching with AI — steal two ideas you can use.

☐ **Fluency** **Explore discipline tools.** For statistics and data science: AI analysis assistants, code copilots for R and Python, AI features in Quarto and Posit.

☐ **Agility** **Keep an experiment log.** One page a month: what worked, what you would teach, and what you would warn students about.

Keep going

Track your progress: tick each item as you complete it, then bring this page to your next department meeting. **Keynote:** <https://www.csueastbay.edu/oaa/files/docs/provost-muscat-b2b-keynote.pdf>



Faculty Checklist

Part B: Bringing AI into your classes · 13 items

Small, deliberate changes that put AI to work in your courses — without handing the learning process to algorithms.

- ☐ **Fluency** **Write your syllabus AI policy.** Name the permitted tools, disclosure and attribution rules, verification expectations, and privacy requirements.
- ☐ **Fluency** **Design a paired assignment.** One AI-allowed version and one AI-free version — students experience both and reflect on the difference.
- ☐ **Fluency** **Require citation and verification.** Students attach the prompt, the tool used, and a note on how they checked the result.
- ☐ **Fluency** **Teach prompting and critique.** A 15–25 minute in-class lab: prompt, compare outputs, identify the errors together.
- ☐ **Fluency** **Start low-stakes.** One small AI micro-activity before graded work — scaffold from low-stakes to high-stakes.
- ☐ **Human-Centric** **Add an oral check.** A short viva: students explain their work and their AI use in their own words.
- ☐ **Human-Centric** **Include an ethics case.** Algorithmic bias, data privacy, disparate impact — students propose mitigations, not just spot problems.
- ☐ **Human-Centric** **Two-audience deliverable.** A technical report plus a one-page brief a non-technical stakeholder can act on.
- ☐ **Human-Centric** **Assign rotating team roles.** Lead, reviewer, communicator — everyone practices leadership and relational skills.
- ☐ **Agility** **Build in a mid-course change.** New data, a shifted requirement, or a broken assumption students must respond to.
- ☐ **Agility** **Run an agentic oversight simulation.** Students set a goal for an AI agent, monitor its output, quality-check it, and decide where to override.
- ☐ **Agility** **Add a real-client deliverable.** A community partner or campus project: AI accelerates, students own the interpretation.
- ☐ **Human-Centric** **End with the Provost's two asks.** Where did AI deepen human interpretation — and where did not using AI teach something?

“If we embed these experiences into our teaching now, we will not just keep pace with the future — we will shape it.”

— Provost Muscat, Back to the Bay 2025